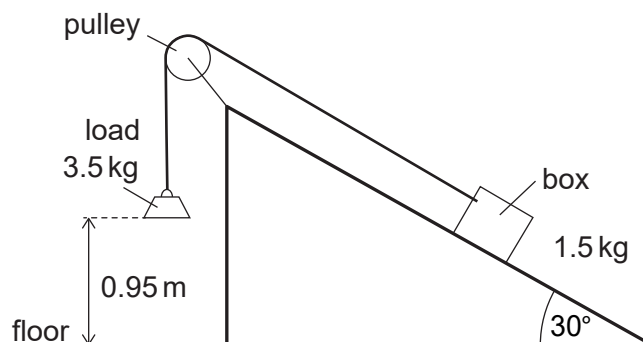


14. A student uses a load to pull a box up a ramp inclined at  $30^\circ$ . A string of constant length and negligible mass connects the box to the load that falls vertically. The string passes over a pulley that runs on a frictionless axle. Friction acts between the base of the box and the ramp. Air resistance is negligible.



The load has a mass of 3.5 kg and is initially 0.95 m above the floor. The mass of the box is 1.5 kg.

The load is released and accelerates downwards.

- (a) Outline **two** differences between the momentum of the box and the momentum of the load at the same instant.

[2]

.....

.....

.....

.....

- (b) The vertical acceleration of the load downwards is  $2.4 \text{ ms}^{-2}$ .

Calculate the tension in the string.

[2]

.....

.....

.....

.....

(This question continues on the following page)

28 April 2022 / TZ1 / HL Paper 2 / original question 1 / 2222-6508

28 April 2022 / TZ1 / HL Paper 2 / original question 1 / 2222-6508

(c) Show that the speed of the load when it hits the floor is about  $2.1 \text{ m s}^{-1}$ . [2]

.....

.....

.....

.....

- (d) After the load has hit the floor, the box travels a further 0.35 m along the ramp before coming to rest. Determine the average frictional force between the box and the surface of the ramp. [4]

[illegible]

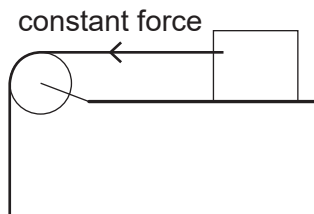
Question 14 / page 2 of 3

## PAPER 2 / QUESTION 14

28 April 2022 / TZ1 / HL Paper 2 / original question 1 / 2222-6508

### (Question 14 continued)

- (e) The student then makes the ramp horizontal and applies a constant horizontal force to the box. The force is just large enough to start the box moving. The force continues to be applied after the box begins to move.



Explain, with reference to the frictional force acting, why the box accelerates once it has started to move.

[3]

.....

.....

.....

.....

.....

.....